Patrick Amadeus Irawan

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Website LinkedIn Github

Research Summary

My research focuses on understanding how multimodal systems fail to reason in real-world settings and further develops post-training methods to improve their performance on fine-grained and complex visual tasks. I have published 10+ papers at CVPR, NAACL, EMNLP, and COLING on vision-language models, with focus on multimodal reasoning, robustness, evaluation, and post-training. I further aim to build multimodal systems that reason effectively across diverse perceptual / action-based inputs without relying on modality-specific shortcuts.

Keywords: computer vision, NLP, vision-language models, unified multimodal models, RL.

Education

Ph.D. in Natural Language Processing — MBZUAI *2025 - 2029*
- Advisors: Alham Fikri Aji and Yova Kementchedjhieva
- *multimodal learning; VLM; LLM; post-training; RL*

Global Study Program — University of California, Davis *2023*
- *computational neuroscience, tech management*

Bachelor in Computer Science — Institut Teknologi Bandung *2020 - 2024*
- *visual question answering; synthetic data generation; reasoning LLM*

Publications

* **First author** † **Second author**

LinguDistill: Recovering Linguistic Ability in Vision Language Models via Selective Cross-Modal Distillation* — *Preprint* 2026
VLM, SFT, post-training, knowledge distillation, cross-modal learning, architecture

Anthropogenic Regional Adaptation in Multimodal Vision-Language Model — *Under Review* 2026
VLM, SFT, post-training, data curation, evaluation, multilingual, multimodal

Vision Language Models are Confused Tourists* — CVPR 2026
VLM, visual reasoning, robustness, hallucination, interpretability

A Massive-Scale Multilingual Multi-Cultural Multimodal RAG† — CVPR 2026
RAG, image retrieval, multilingual, multimodal, multicultural, test-time scaling

Counting to Four is Still a Chore for VLMs* — MMFM @ CVPR 2026
VLM, numerical reasoning, evaluation, interpretability, object detection, RL

Entropy2Vec: Crosslingual Language Modeling Entropy as End-to-End Learnable Language Representations* — MRL @ EMNLP 2025
LLM, cross-lingual, entropy, representation learning, language modeling, unsupervised learning

WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural VQA on Global Cuisines* — NAACL 2025 (Best Theme Paper 🏆)
VQA, multilingual, multicultural, benchmark, large-scale evaluation, data curation

Seeing Culture: A Benchmark for Visual Reasoning and Grounding† — EMNLP 2025
visual reasoning, visual grounding, benchmark, evaluation, object detection, multicultural

Predicting Language Model Performance on Multilingual Tasks via Proxy Models — NAACL 2025
multilingual, evaluation, proxy models, regression, SFT

Towards Efficient and Robust VQA-NLE Data Generation with Large Vision-Language Models* — COLING 2025
VQA, VLM, NLU, NLG, synthetic data generation, visual reasoning

SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages — EMNLP 2024
multilingual, multicultural, video understanding, speech understanding, evaluation

Work Experience

Research Engineer *Singapore*
Singapore Management University ***Feb 2025 – Sep 2025***

- Led multimodal reasoning evaluation across 10K+ image–text samples, coordinating 5 researchers (juniors, externals), resulting in EMNLP main-track publication on VLM robustness for knowledge-intensive VQA and visual grounding.
- Designed and executed end-to-end data curation pipelines (data collection via crowdsourcing, filtering, packing), reducing experiment turnaround time and improving benchmarking coverage by 50%.

Software Engineer *Germany*
IT Bauschmiede ***Nov 2024 – Jul 2025***

- Part of core logistics team optimizing real-time scheduling, invoicing, and geolocation systems serving 5+ enterprise clients, reducing manual operations by ∼4%
- Architected microservices migration from monolithic MVC, reducing system latency by ∼20% and improving system uptime and scalability

Data Scientist Intern
Supertype ***2023***

- Built predictive models reducing churn rate by 12% by replacing neural approaches with foundational statistical models. Improved model explainability and stakeholder adoption through feature-driven modeling.

Software Engineer Intern
Blibli ***2022***

- Scaled full-stack checkout and transaction testing coverage by 5% out of total volume over 6 months. Contributed majority on redundant SQL migration jobs project, reducing data pipeline overhead by 8% across services.

Software Engineer Intern
Ruangguru ***2022***

- Architected and scaled internal autograder system supporting 1K+ students/week, improving grading efficiency by ∼70%

Achievement & Awards

OpenAI Micro Grant Award **Grantee**	*2025*
NAACL Best Theme Paper Award **Winner**	*2025*
BCG SEA Emeralds Case Competition **Winner**	*2024*
Indonesian International Student Mobility Awards @ UC Davis **Awardee**	*2023*
DOMO Higher Education Case Competition **Finalist**	*2023*
CSLeaders Full Bachelor Tuition Scholarship **Awardee**	*2022*
Gemastik XV Data Mining Division **Finalist**	*2022*

Technical Skills

Programming: Python, Java, C++, Typescript, SQL, Bash
ML / AI Frameworks: PyTorch, TensorFlow, Hugging Face, CUDA, OpenCV, Weights & Biases (wandb)
Tools & Systems: LangChain, LangSmith, distributed training, microservices, APIs, Agents
Multimodal & NLP: Vision-Language Models (VLMs), Large Language Models (LLMs), Unified Multimodal Models (UMMs), multimodal learning, cross-modal learning, SFT, RL, RAG, knowledge distillation
Research Areas: deep learning, computer vision, visual reasoning, video understanding, multilingual modeling, synthetic data generation, large-scale evaluations, post-training
Other Research Skills: experiment design, grant writing, scientific communication, scientific mentoring, project management